

Лабораторная работа №15

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Цель работы

Приобретение практических навыков администрирования сетевых подсистем

На сервере создадим файл конфигурации сетевого хранения журналов в каталоге `/etc/rsyslog.d` (рис. @fig-1):

В файле конфигурации `/etc/rsyslog.d/netlog-server.conf` включим приём записей журнала по TCP-порту 514 (рис. @fig-2):

Перезапустим службу `rsyslog` и посмотрим, какие порты, связанные с `rsyslog`, прослушиваются (рис. @fig-3):

На сервере настроим межсетевой экран для приёма сообщений по TCP-порту 514 (рис. @fig-4):

На клиенте создадим файл конфигурации сетевого хранения журналов (рис. @fig-5):

В файле конфигурации `/etc/rsyslog.d/netlog-client.conf` включим перенаправление сообщений журнала на 514 TCP-порт сервера (рис. @fig-6):

Перезапустим службу `rsyslog` на клиенте для применения изменений (рис. @fig-7):

```
(gedit:9739): dconf-WARNING **: 17:32:11.810: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server ~]# systemctl restart rsyslog
[root@server ~]# lsof | grep TCP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1000/doc
Output information may be incomplete.
systemd      1                root    53u      IPv4        46424      0t0      T
CP *:sunrpc (LISTEN)
systemd      1                root    55u      IPv6        45670      0t0      T
CP *:sunrpc (LISTEN)
cupsd        953              root     6u      IPv6        21155      0t0      T
CP localhost:ipp (LISTEN)
cupsd        953              root     7u      IPv4        21156      0t0      T
CP localhost:ipp (LISTEN)
sshd         968              root     3u      IPv4        21197      0t0      T
CP *:down (LISTEN)
sshd         968              root     4u      IPv6        21199      0t0      T
CP *:down (LISTEN)
sshd         968              root     5u      IPv4        21201      0t0      T
CP *:ssh (LISTEN)
sshd         968              root     6u      IPv6        21203      0t0      T
CP *:ssh (LISTEN)
mariadb      1060             mysql    20u      IPv6        23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1149 mariadb      mysql    20u      IPv6        23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1215 mariadb      mysql    20u      IPv6        23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1233 mariadb      mysql    20u      IPv6        23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1238 mariadb      mysql    20u      IPv6        23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1349 mariadb      mysql    20u      IPv6        23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 1352 mariadb      mysql    20u      IPv6        23706      0t0      T
CP *:mysql (LISTEN)
mariadb      1060 9525 mariadb      mysql    20u      IPv6        23706      0t0      T
CP *:mysql (LISTEN)
firefox      2784             vagrant  55u      IPv4        177482     0t0      T
CP server.user.net:50258->146.75.121.91:https (ESTABLISHED)
firefox      2784             vagrant  58u      IPv4        174221     0t0      T
```

Рис. 1: Этап 1



Рис. 2: Этап 2

```
root@client:/etc/rsyslog.d

[vagrant@client ~]$ sudo -i
[root@client ~]# cd /etc/rsyslog.d
[root@client rsyslog.d]# touch netlog-client.conf
[root@client rsyslog.d]# gedit /etc/rsyslog.d/netlog-client.conf

(gedit:5331): dconf-WARNING **: 17:46:55.768: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)

(gedit:5331): dconf-WARNING **: 17:46:55.776: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)

(gedit:5331): dconf-WARNING **: 17:46:56.122: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)

(gedit:5331): dconf-WARNING **: 17:46:56.129: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)

(gedit:5331): dconf-WARNING **: 17:46:56.131: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)

** (gedit:5331): WARNING **: 17:47:09.058: Set document metadata failed: Setting attribute metadata::gedit-spell
-language not supported

** (gedit:5331): WARNING **: 17:47:09.059: Set document metadata failed: Setting attribute metadata::gedit-encod
ing not supported

** (gedit:5331): WARNING **: 17:47:15.649: Set document metadata failed: Setting attribute metadata::gedit-spell
-language not supported

** (gedit:5331): WARNING **: 17:47:15.652: Set document metadata failed: Setting attribute metadata::gedit-encod
ing not supported

** (gedit:5331): WARNING **: 17:47:16.810: Set document metadata failed: Setting attribute metadata::gedit-posit
ion not supported

(gedit:5331): dconf-WARNING **: 17:47:16.831: failed to commit changes to dconf: Failed to execute child process
"dbus-launch" (No such file or directory)
[root@client rsyslog.d]# systemctl restart rsyslog
[root@client rsyslog.d]#
```

Рис. 3: Этап 3

```
[root@server ~]# tail -f /var/log/messages
Mar 12 17:43:08 server rsyslogd[1325]: [origin software="rsyslogd" swVersion="8.2310.0-4.el9" x-p
id="1325" x-info="https://www.rsyslog.com"] exiting on signal 15.
Mar 12 17:43:08 server systemd[1]: rsyslog.service: Deactivated successfully.
Mar 12 17:43:08 server systemd[1]: Stopped System Logging Service.
Mar 12 17:43:08 server systemd[1]: rsyslog.service: Consumed 3.382s CPU time.
Mar 12 17:43:08 server systemd[1]: Starting System Logging Service...
Mar 12 17:43:08 server rsyslogd[9790]: error querying port '**UNSPECIFIED**': Servname not suppor
ted for ai_socktype [v8.2310.0-4.el9 try https://www.rsyslog.com/e/2076 ]
Mar 12 17:43:08 server rsyslogd[9790]: Could not create tcp listener, ignoring port 514S bind-add
ress **UNSPECIFIED**. [v8.2310.0-4.el9 try https://www.rsyslog.com/e/2076 ]
Mar 12 17:43:08 server systemd[1]: Started System Logging Service.
Mar 12 17:43:08 server rsyslogd[9790]: [origin software="rsyslogd" swVersion="8.2310.0-4.el9" x-p
id="9790" x-info="https://www.rsyslog.com"] start
Mar 12 17:43:08 server rsyslogd[9790]: imjournal: journal files changed, reloading... [v8.2310.0
-4.el9 try https://www.rsyslog.com/e/0 ]
```

Рис. 4: Этап 4

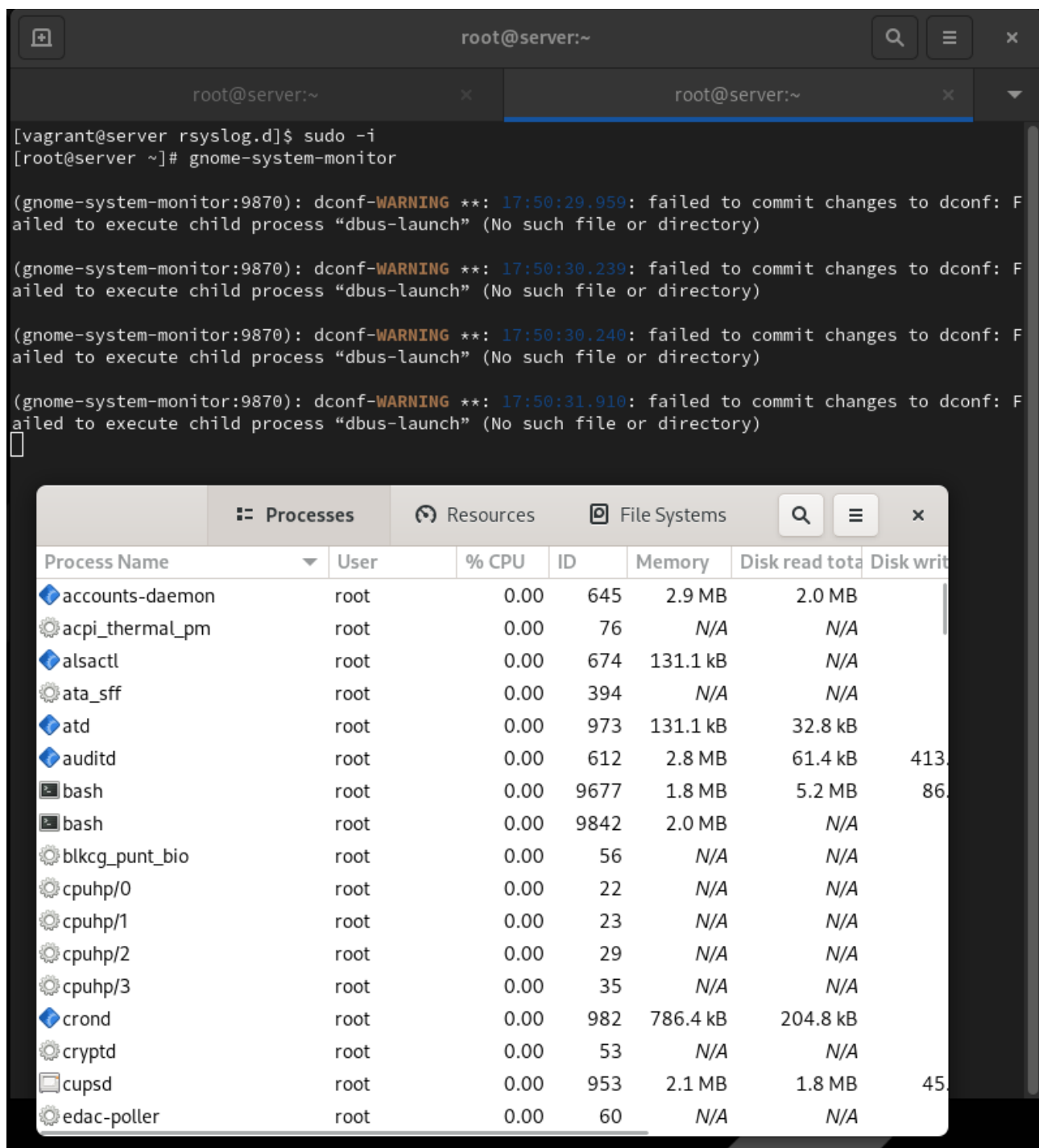


Рис. 5: Этап 5

```
root@server:~
[vagrant@server rsyslog.d]$ sudo -i
[root@server ~]# dnf -y install lnav
Last metadata expiration check: 1:59:28 ago on Thu 12 Mar 2026 03:52:25 PM UTC.
Dependencies resolved.
=====
Package                Architecture      Version           Repository        Size
=====
Installing:
lnav                    x86_64            0.11.1-1.el9     epel               2.4 M
Transaction Summary
=====
Install 1 Package

Total download size: 2.4 M
Installed size: 6.1 M
Downloading Packages:
lnav-0.11.1-1.el9.x86_64.rpm                                1.8 MB/s | 2.4 MB    00:01
-----
Total                                                         1.1 MB/s | 2.4 MB    00:02
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : lnav-0.11.1-1.el9.x86_64      1/1
  Running scriptlet: lnav-0.11.1-1.el9.x86_64    1/1
  Verifying      : lnav-0.11.1-1.el9.x86_64      1/1

Installed:
  lnav-0.11.1-1.el9.x86_64

Complete!
[root@server ~]#
```

Рис. 6: Этап 6

```
2026-03-12T17:53:14 UTC Press ENTER to focus on the breadcrumb bar
LOG >2026-03-12T17:31:22.000>syslog_log>messages[16,465]>gnome-shell[2083]>
Mar 12 17:31:22 server gnome-shell[2083]: Window manager warning: Buggy client sent a _NET_ACTI
Mar 12 17:31:23 server systemd[1]: systemd-hostnamed.service: Deactivated successfully.
Mar 12 17:31:45 server gnome-shell[2083]: Window manager warning: Buggy client sent a _NET_ACTI
Mar 12 17:33:21 server chronyd[658]: Detected falseticker 195.122.253.38 (2.rocky.pool.ntp.org)
Mar 12 17:43:08 server systemd[1]: Stopping System Logging Service...
Mar 12 17:43:08 server rsyslogd[1325]: [origin software="rsyslogd" swVersion="8.2310.0-4.el9" x
Mar 12 17:43:08 server systemd[1]: rsyslog.service: Deactivated successfully.
Mar 12 17:43:08 server systemd[1]: Stopped System Logging Service.
Mar 12 17:43:08 server systemd[1]: rsyslog.service: Consumed 3.382s CPU time.
Mar 12 17:43:08 server systemd[1]: Starting System Logging Service...
Mar 12 17:43:08 server rsyslogd[9790]: error querying port '**UNSPECIFIED**': Servname not supp
Mar 12 17:43:08 server rsyslogd[9790]: Could not create tcp listener, ignoring port 514S bind-a
Mar 12 17:43:08 server systemd[1]: Started System Logging Service.
Mar 12 17:43:08 server rsyslogd[9790]: [origin software="rsyslogd" swVersion="8.2310.0-4.el9" x
Mar 12 17:43:08 server rsyslogd[9790]: imjournal: journal files changed, reloading... [v8.2310
Mar 12 17:50:03 server systemd[1906]: Started VTE child process 9814 launched by gnome-terminal
Mar 12 17:50:07 server systemd[1]: Starting Hostname Service...
Mar 12 17:50:07 server systemd[1]: Started Hostname Service.
Mar 12 17:50:37 server systemd[1]: systemd-hostnamed.service: Deactivated successfully.
Mar 12 17:51:38 server systemd[1906]: Started VTE child process 9893 launched by gnome-terminal
Mar 12 17:51:41 server systemd[1]: Starting Hostname Service...
Mar 12 17:51:41 server systemd[1]: Started Hostname Service.
Mar 12 17:51:58 server systemd[1]: Started /usr/bin/systemctl start man-db-cache-update.
Mar 12 17:51:58 server systemd[1]: Starting man-db-cache-update.service...
Mar 12 17:52:02 server systemd[1]: man-db-cache-update.service: Deactivated successfully.
Mar 12 17:52:02 server systemd[1]: Finished man-db-cache-update.service.
Mar 12 17:52:02 server systemd[1]: man-db-cache-update.service: Consumed 1.657s CPU time.
Mar 12 17:52:02 server systemd[1]: run-rc686ad34f9c549e8bd890b834106c10d.service: Deactivated s
Mar 12 17:52:04 server systemd[1]: Starting PackageKit Daemon...
Mar 12 17:52:04 server systemd[1]: Started PackageKit Daemon.
Mar 12 17:52:11 server systemd[1]: systemd-hostnamed.service: Deactivated successfully.
```

Files :: Text Filters :: Press TAB to edit
L16,465 100% ? :View Help
Press e/E to move forward/backward through error messages

Рис. 7: Этап 7

```
[vagrant@server rsyslog.d]$ sudo -i
[root@server ~]# cd /vagrant/provision/server
[root@server server]# mkdir -p /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server server]# cp -R /etc/rsyslog.d/netlog-server.conf /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server server]# cd /vagrant/provision/server
[root@server server]# touch netlog.sh
[root@server server]# chmod +x netlog.sh
[root@server server]# gedit netlog.sh

(gedit:22854): dconf-WARNING **: 19:17:52.030: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:22854): dconf-WARNING **: 19:17:52.040: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:22854): dconf-WARNING **: 19:17:53.651: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:22854): dconf-WARNING **: 19:17:53.674: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:22854): dconf-WARNING **: 19:17:53.674: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
```

Рис. 8: Этап 8

На сервере посмотрим один из файлов журнала для проверки сбора сообщений (рис. @fig-8):

На сервере под пользователем запустим графическую программу для просмотра журналов gnome-system-monitor (рис. @fig-9):

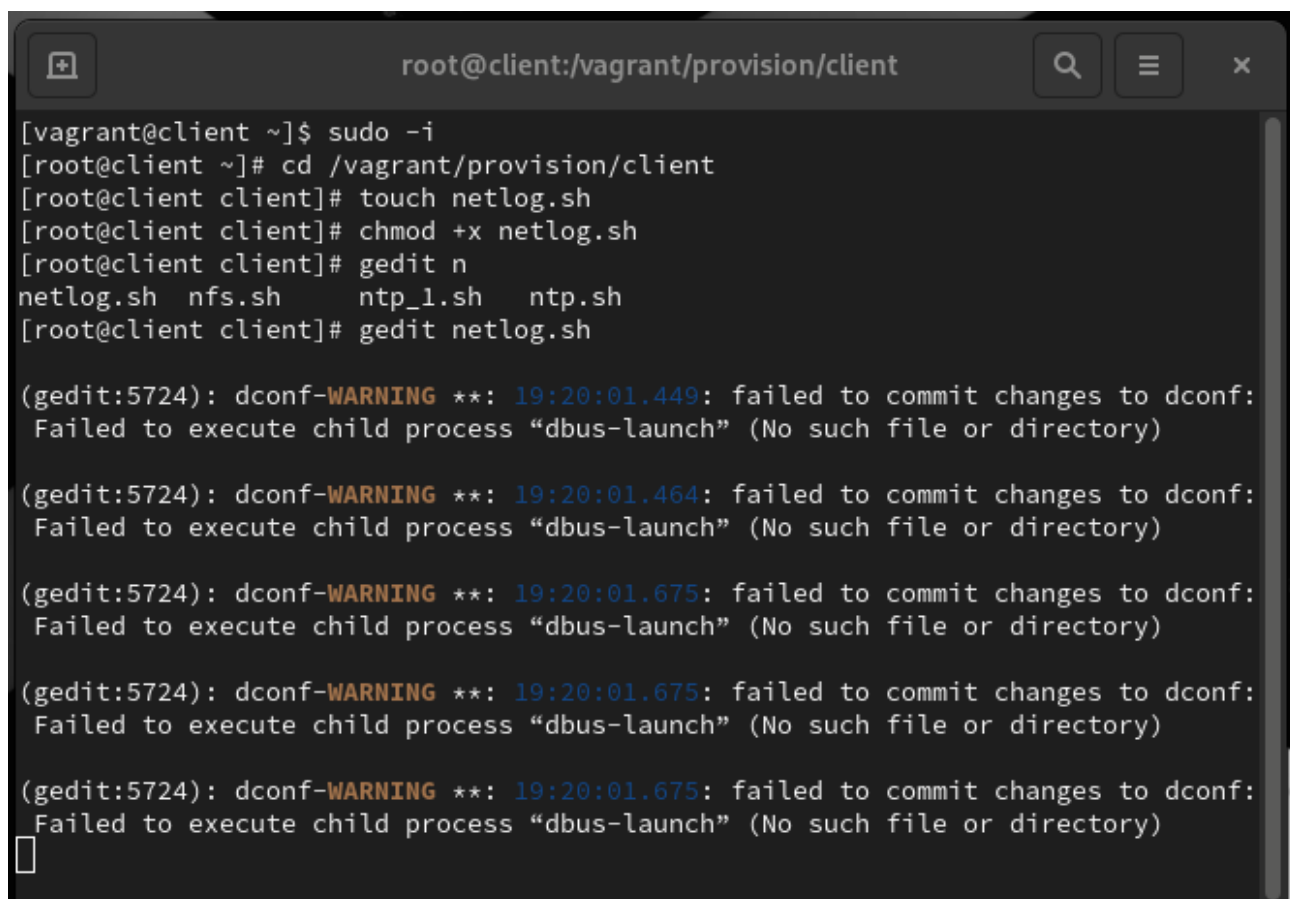


```
netlog.sh
/vagrant/provision/server

1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Copy configuration files"
4 cp -R /vagrant/provision/server/netlog/etc/* /etc
5 restorecon -vR /etc
6 echo "Configure firewall"
7 firewall-cmd --add-port=514/tcp
8 firewall-cmd --add-port=514/tcp --permanent
9 echo "Start rsyslog service"
10 systemctl restart rsyslog
```

Рис. 9: Этап 9

На сервере установим просмотрщик журналов системных сообщений Inav (рис. @fig-10):



```
root@client:/vagrant/provision/client

[vagrant@client ~]$ sudo -i
[root@client ~]# cd /vagrant/provision/client
[root@client client]# touch netlog.sh
[root@client client]# chmod +x netlog.sh
[root@client client]# gedit n
netlog.sh  nfs.sh      ntp_1.sh  ntp.sh
[root@client client]# gedit netlog.sh

(gedit:5724): dconf-WARNING **: 19:20:01.449: failed to commit changes to dconf:
Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:5724): dconf-WARNING **: 19:20:01.464: failed to commit changes to dconf:
Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:5724): dconf-WARNING **: 19:20:01.675: failed to commit changes to dconf:
Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:5724): dconf-WARNING **: 19:20:01.675: failed to commit changes to dconf:
Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:5724): dconf-WARNING **: 19:20:01.675: failed to commit changes to dconf:
Failed to execute child process "dbus-launch" (No such file or directory)
```

Рис. 10: Этап 10

Просмотрим логи с помощью lnav для удобного анализа системных сообщений (рис. @fig-11):



```
*netlog.sh
/vagrant/provision/client

1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Install needed packages"
4 dnf -y install lnav
5 echo "Copy configuration files"
6 cp -R /vagrant/provision/client/netlog/etc/* /etc
7 restorecon -vR /etc
8 echo "Start rsyslog service"
9 systemctl restart rsyslog
```

Рис. 11: Этап 11

Выводы

В ходе выполнения лабораторной работы были приобретены практические навыки

Спасибо за внимание!

Вопросы?